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Force of Attraction between Two Electric Charges and Two Parallel Conductors Separated by Varying Distance

Aim:

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB. Objectives:

1. To compute the electrostatic force of attraction or repulsion between two point charges using Coulomb's law.

2. To compute the magnetic force per unit length between two current-carrying parallel conductors.

3. To analyze and plot the variation of force with separation distance and charge or current magnitude.

Test case (i) Determination and plotting of force between two electric charges with varying distance

```
clc;

clear;

q1 = input('Enter the charge q1=');

q2 = input('Enter the charge q2=');

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.1,10,100);

F = k*q1*q2./(r.^2);

disp('Force at r = 1 m is:');

disp(k*q1*q2/(1^2));

plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two electric charges');
```

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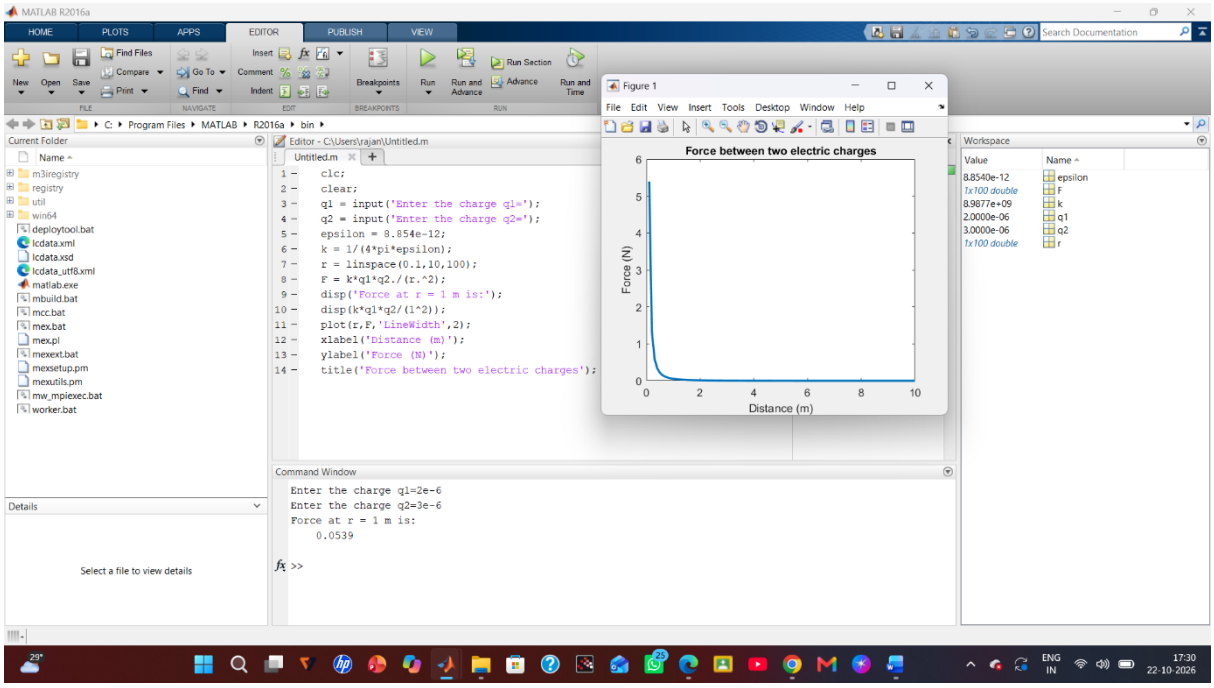
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Output:-

Enter the charge q1=2e-6

Enter the charge q2=3e-6

Force at r = 1 m is: 0.0539 N



Test case (ii) Determination and plotting of force between two electric charges with different magnitudes

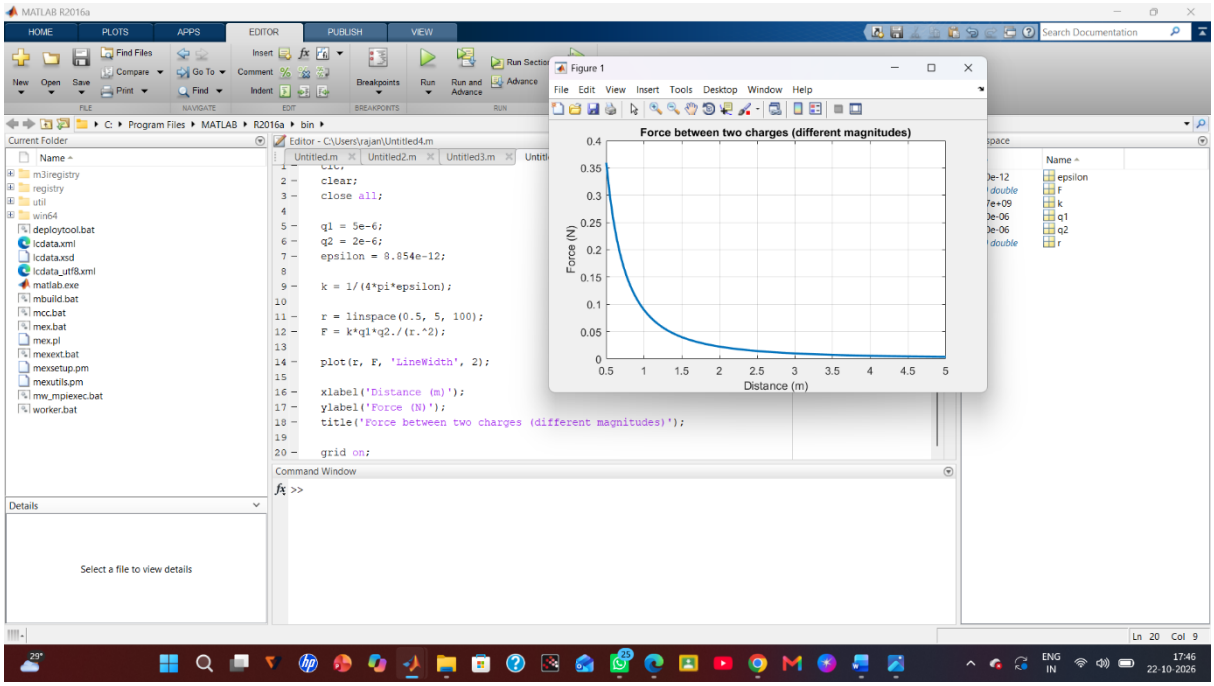
```
clc;
q1 = 5e-6;
q2 = 2e-6;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
r = linspace(0.5,5,100);
F = k*q1*q2./(r.^2);
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
```

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ylabel('Force (N)');

title('Force between two charges (different magnitudes)');



Test case (iii) Determination and plotting of force per unit length between two parallel conductors with varying distance

clc;

mu0 = 4*pi*1e-7;

I1 = 5;

I2 = 10;

d = linspace(0.1,10,100);

F = (mu0*I1*I2)./(2*pi*d);

plot(d,F,'LineWidth',2);

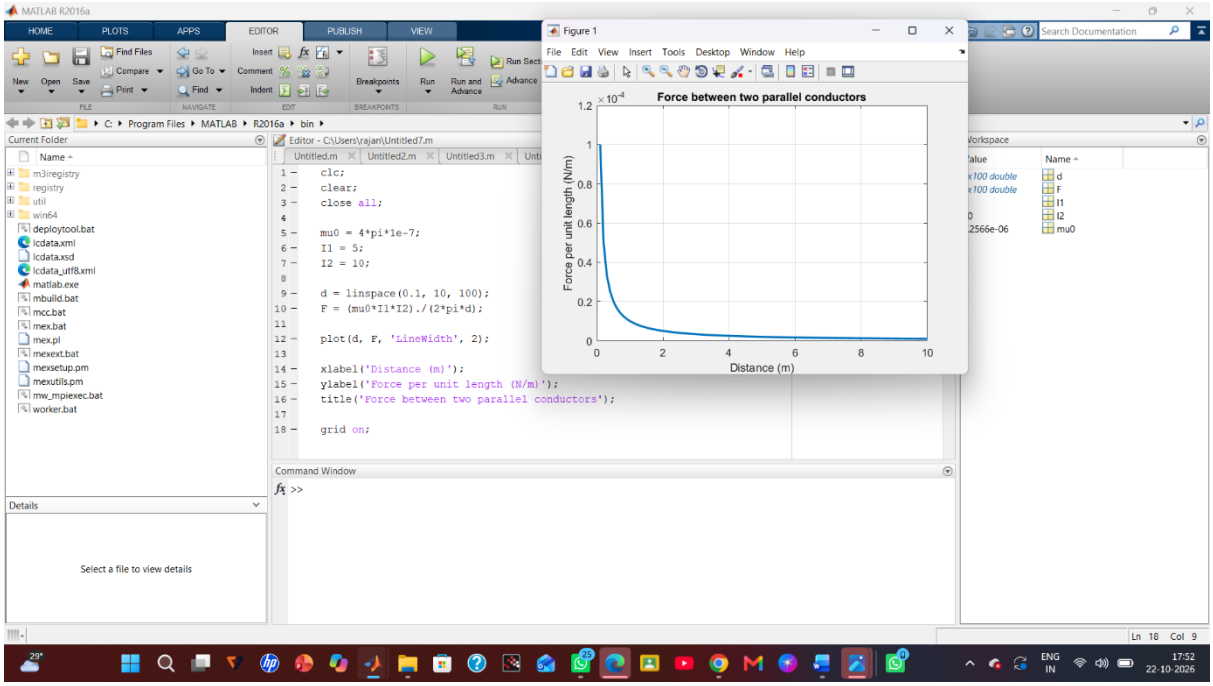
xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between two parallel conductors');

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Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

```
clc;

clear;

close all;


mu0 = 4*pi*1e-7;

I1 = 5;

I2 = 10;


d = linspace(0.1, 10, 100);

F = (mu0*I1*I2)./(2*pi*d);


plot(d, F, 'LineWidth', 2);
```

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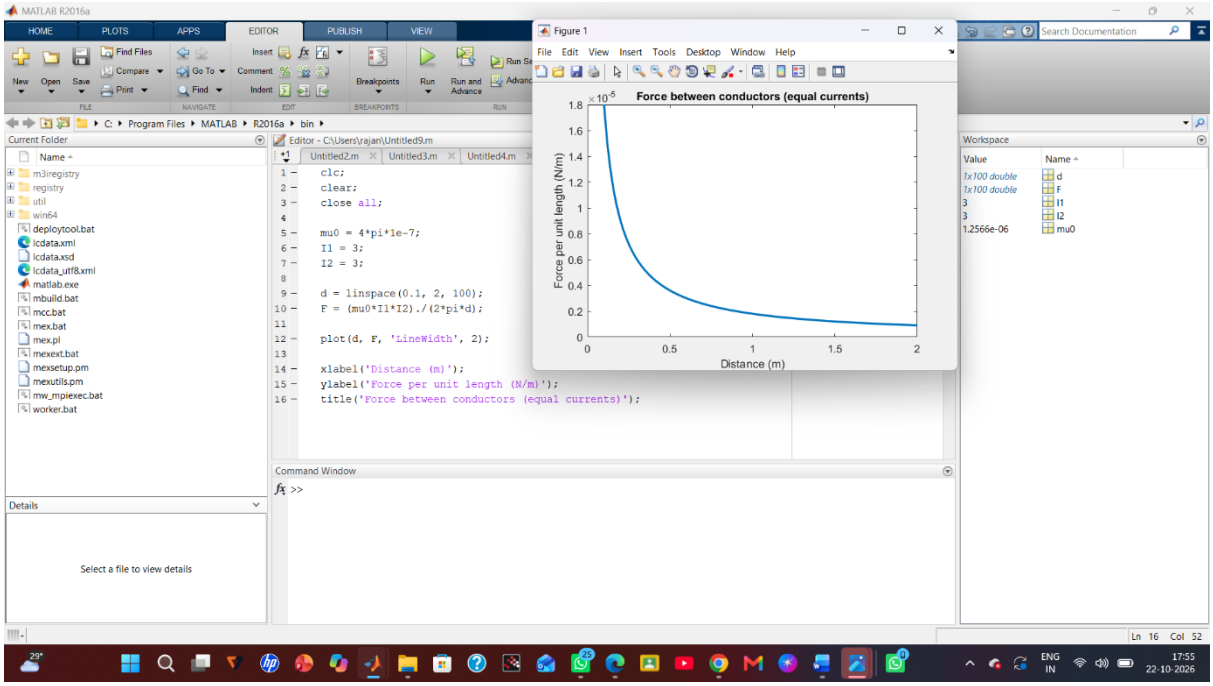
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```
xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between two parallel conductors');


grid on;
```



Test case (v) : Magnetostatic Analysis – Biot–Savart / Ampere-Law Field Plot

```
%% Biot-Savart / Ampere's Law Magnetic Field Analysis

% Magnetic field around a long straight current-carrying conductor

clear; clc; close all;


%% 1. Input Parameters & Constants

I     = 5;                % Current in Amperes

mur   = 1;                % Relative permeability

mu0   = 4 * pi * 1e-7;    % Permeability of free space (H/m)
```

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```
mu     = mu0 * mur;          % Absolute permeability

% Radial distance vector from conductor
r = linspace(0.01, 1.00, 100); % Distance in meters (1 cm to 1 m)

%% 2. Analytical Formulation (Ampere's Law)
H = I ./ (2 * pi * r);        % Magnetic Field Intensity (A/m)
B = mu .* H;                  % Magnetic Flux Density (T)

%% 3. Independent Numerical Validation (Biot-Savart Law)
% Finite wire of length 100 m (-50 m to +50 m)
% Using vectorized Biot-Savart integration over the subtended angle alpha
compute_H_bs = @(dist) ...
    (I / (4 * pi * dist)) * trapz(linspace(-atan(50/dist), atan(50/dist), 20001), ...
                                  cos(linspace(-atan(50/dist), atan(50/dist), 20001)));

H_num = arrayfun(compute_H_bs, r);
B_num = mu .* H_num;

% Percentage deviation vs. infinite wire analytical solution
H_error = abs((H_num - H) ./ H) * 100;
B_error = abs((B_num - B) ./ B) * 100;

%% 4. Tabular Output for Selected Distances
```

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```
r_selected = [0.01, 0.05, 0.10, 0.20, 0.50, 1.00];


fprintf('\n');

fprintf('%s\n', repmat('-', 1, 65));

fprintf(' MAGNETIC FIELD AROUND LONG STRAIGHT CONDUCTOR\n');

fprintf('%s\n', repmat('-', 1, 65));

fprintf('Current I = %.2f A\n', I);

fprintf('Relative permeability mur = %.2f\n', mur);

fprintf('%s\n', repmat('-', 1, 65));

fprintf('\n%8s %12s %12s %14s %14s\n', 'r (m)', 'H (A/m)', 'B (T)', 'H Error (%)', 'B Error (%)');

fprintf('%s\n', repmat('-', 1, 65));


for r_val = r_selected
    % Locate closest index in evaluation vector
    [~, idx] = min(abs(r - r_val));


    H_val = I / (2 * pi * r_val);
    B_val = mu * H_val;


    fprintf('%8.2f %12.4f %12.4e %14.4f %14.4f\n', ...
            r_val, H_val, B_val, H_error(idx), B_error(idx));
end
```

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```
%% 5. Maximum Deviation Summary & Results Statement
max_H_error = max(H_error);
max_B_error = max(B_error);

fprintf('\nMaximum percentage deviation in H = %.6f %%\n', max_H_error);
fprintf('Maximum percentage deviation in B = %.6f %%\n', max_B_error);

fprintf('\nRESULT:\n');
fprintf('The magnetic field intensity H and magnetic flux density B\n');
fprintf('were calculated and plotted using MATLAB.\n');
fprintf('Both H and B decrease approximately as 1/r.\n');
fprintf('The MATLAB results agree with the analytical expressions.\n');

%% 6. Visualization & Plots
% Figure 1: H vs r
figure('Name', 'Magnetic Field Intensity vs Distance', 'Color', 'w');
plot(r, H, 'LineWidth', 2);
grid on;
xlabel('Radial Distance r (m)');
ylabel('Magnetic Field Intensity H (A/m)');
title('Magnetic Field Intensity H versus Radial Distance');

% Figure 2: B vs r
figure('Name', 'Magnetic Flux Density vs Distance', 'Color', 'w');
```

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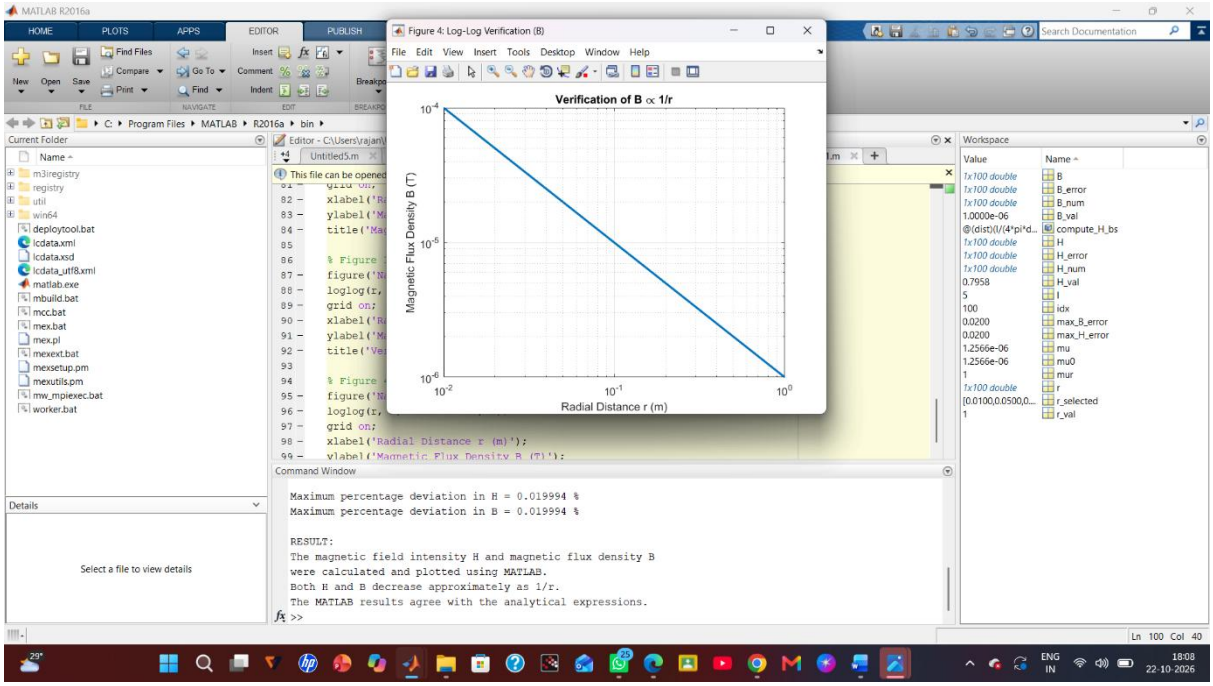
```
plot(r, B, 'LineWidth', 2);
grid on;
xlabel('Radial Distance r (m)');
ylabel('Magnetic Flux Density B (T)');
title('Magnetic Flux Density B versus Radial Distance');

% Figure 3: Log-Log Verification for H
figure('Name', 'Log-Log Verification (H)', 'Color', 'w');
loglog(r, H, 'LineWidth', 2);
grid on;
xlabel('Radial Distance r (m)');
ylabel('Magnetic Field Intensity H (A/m)');
title('Verification of H \propto 1/r');

% Figure 4: Log-Log Verification for B
figure('Name', 'Log-Log Verification (B)', 'Color', 'w');
loglog(r, B, 'LineWidth', 2);
grid on;
xlabel('Radial Distance r (m)');
ylabel('Magnetic Flux Density B (T)');
title('Verification of B \propto 1/r');
```

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S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifie s most quantities and models correctly	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel current force calculations	Calculat ions are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25

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			correctly with units.				
3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissions	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30
4	Explain and report results	Explain and report results	Explains attraction/repulsion and reports assumptions, units and conclusions clearly.	Conclusion is correct with minor documentation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20
Total							100